Lecture 14: Wave-Particle Duality, Photoelectric Effect, Compton Effect.

I. REVISITING THE DOUBLE SLIT EXPERIMENT.

For a long time, the interference of light was regarded as a decisive victory for the wave theory. When a coherent beam passes through a pair of slits and forms a pattern of alternating bright and dark fringes on a screen, it is almost impossible not to think of it as a wave phenomenon. The superposition principle explains it: where crests meet crests, the intensity is high; where crests meet troughs, the waves cancel.

Now imagine repeating the same double-slit experiment, but gradually dimming the source. At full brightness, the interference pattern is sharp and vivid. When the light is made weaker, the same pattern appears, but fainter. So far, nothing unexpected. But then, let the source become extremely dim, so weak that only a few photons reach the screen in an entire night. The next morning, instead of a pattern of continuous light and dark bands, you see only a handful of random dots—tiny exposed spots on the photographic plate (See Fig. 1).

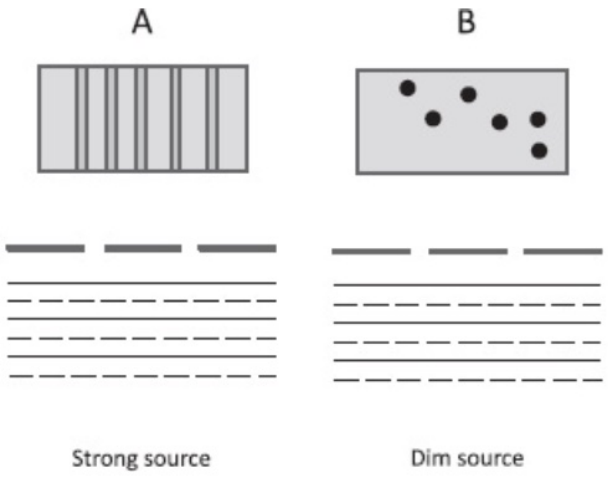


FIG. 1.

If the intensity is reduced even further, you might find that only a single dot appears

after a long time. This is deeply puzzling. If light were a continuous wave, no matter how faint, it should gently illuminate the entire screen. It cannot choose one pixel and ignore the rest. Clearly, something localized is arriving at the detector. Light, it seems, is made of individual quanta that deliver energy in discrete lumps.

When one measures the momentum transferred to the film at each impact, something remarkable appears. Every exposed pixel receives exactly the same amount of momentum, regardless of where it lands. By varying the wavelength λ of the incident light, one finds that the magnitude of this momentum is related to the wavelength by

$$p = \frac{h}{\lambda} = \hbar k, \tag{1}$$

where h is Planck's constant and $k = 2\pi/\lambda$ is the wave number. The same experiment also shows that each quantum delivers a fixed amount of energy,

$$E = \hbar\omega, \tag{2}$$

where ω is the angular frequency of the light.

It is then natural to describe light of frequency ω and wave number k as being composed of particles—photons—each carrying energy $E = \hbar\omega$ and momentum $p = \hbar k$. Since electromagnetic waves satisfy $\omega = kc$, it follows that

$$E = pc, \tag{3}$$

or equivalently,

$$E^2 = p^2c^2. \tag{4}$$

Comparing this with the relativistic energy–momentum relation,

$$E^2 = p^2c^2 + m^2c^4, \tag{5}$$

we are led to the conclusion that the photon's rest mass must be zero. It has momentum despite being massless because it always moves at the speed of light.

If we now keep the faint source on for many days, these seemingly random spots begin to accumulate. Gradually, they fill out into the same interference pattern that appeared instantly when the source was bright. This slow build-up reveals something profound: light, which behaves like a continuous wave when bright, is actually composed of discrete quanta

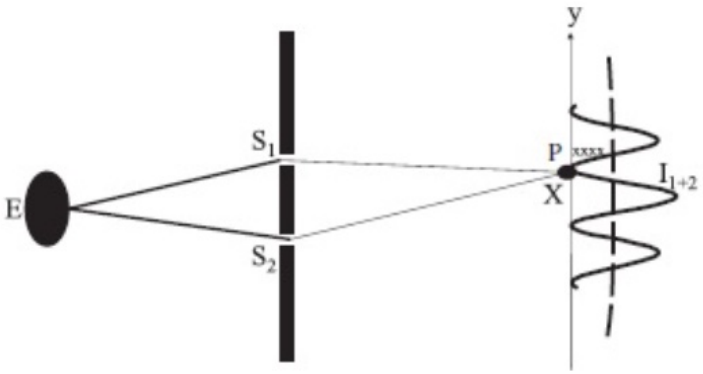


FIG. 2.

that arrive one by one. At high intensity, the vast number of photons creates the illusion of a continuous wave.

At first glance, there is nothing outrageous about this. After all, we already know that materials that look continuous, like water, are composed of molecules. But here the twist is deeper. These photons are not ordinary particles obeying Newton's or Einstein's mechanics. Classical particles travel along definite trajectories and respond to forces in predictable ways. If photons were classical, they would behave in a very different manner in the double-slit experiment.

Consider the two-slit arrangement again. When only one slit, say S_1, is open, the photons travel through it and hit the screen, forming a single-bright pattern. If only S_2 is open, a similar pattern forms elsewhere. Now, if both slits are open, one might expect the total intensity to be simply the sum of the two individual patterns:

$$I_{1+2} = I_1 + I_2. \tag{6}$$

This must be true for classical particles, because a photon that passes through one slit cannot possibly know whether the other slit is open or closed. The number arriving at any point on the screen should therefore be the sum of those coming through each slit independently.

However, this is not what nature does. Experimentally, we find regions where I_{1+2} is smaller than either I_1 or I_2, and in some places even zero. At such points, the two contributions somehow cancel. How can positive numbers of particles cancel one another? This is impossible in a classical framework. Only a wave can interfere destructively. But a wave, as we just saw, cannot explain the localized impacts of individual photons.

We are forced to accept that light is neither purely a wave nor purely a particle. Each photon behaves as a particle when it arrives at the detector, depositing its full energy and momentum at one location. Yet, the probability of where it will land is governed by a wave that spreads through both slits and interferes with itself. When many photons are sent one after another, the accumulated pattern mirrors the intensity distribution of that underlying wave.

To answer the question whether light *is* a particle or a wave, the language that we human beings use is not adequate to describe it. It it is both, it is what it is. The wave guides probabilities; the particle delivers the actual event.

Imagine that a several photons have already formed a beautiful interference pattern on the screen. If we now send in one more photon, there is no way to predict precisely where it will land. We can only say that the probability of finding it at position $\mathbf{r}$ is proportional to the interference pattern on the screen, more chance that it will land in the regions where there are interference maximas. Where the interference is constructive, the probability is high; where it is destructive, the probability is zero. The wave determines the probabilities.

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II. MATTER WAVES

The next major idea came from the French physicist Louis de Broglie. His remarkable idea was simple but revolutionary: if light, once thought to be a pure wave, exhibits particle-like behavior in the form of photons, then perhaps matter, traditionally understood as made of particles should also have wave-like properties.

De Broglie proposed that every particle with momentum p has an associated wavelength given by

$$\lambda = \frac{h}{p} = \frac{2\pi\hbar}{p}. \tag{7}$$

This wavelength, known as the *de Broglie wavelength*, implies that material particles such as electrons, protons, or neutrons should, in principle, exhibit interference just like light waves.

In the case of light, the wavelength λ was a natural and familiar concept, while the photon's discrete momentum p was the surprise. For electrons, the situation is reversed: the momentum p is an obvious, measurable quantity, but the existence of an associated wavelength is the truly unexpected part.

To test de Broglie's hypothesis, experimentalists designed a double-slit experiment for electrons, closely resembling the optical version but with some practical differences. Electrons are emitted from a source—say, a heated electrode that releases them with very small kinetic energy—and then accelerated through a potential difference V. This gives them a momentum

$$p = \sqrt{2mK} = \sqrt{2meV}, \tag{8}$$

where $K = eV$ is the kinetic energy acquired by the electron after acceleration. (Here we use the non-relativistic approximation, valid except for light itself, which always satisfies $E = pc$.)

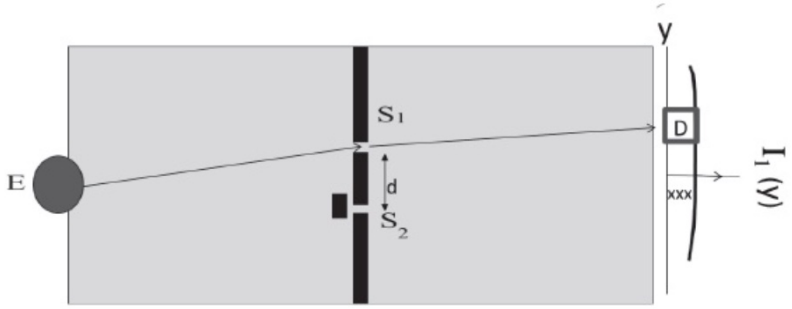


FIG. 3. Top: Electrons travel from the emitter E to a movable detector D with only slit S_1 open. The classical trajectory is shown schematically. The histogram $I_1(y)$ is obtained by counting electron arrivals (marked as crosses) over a fixed time interval.

To detect electrons, the photographic film used for light experiments is replaced by a row of sensitive electron detectors, or a single movable detector that can scan the detection plane (see Fig. 3). Each time an electron hits, the detector registers a count, producing a histogram of arrivals as a function of position.

With only slit S_1 open, one obtains an intensity distribution $I_1(y)$; with only slit S_2 open, a similar pattern $I_2(y)$ appears. When both slits are open, the combined pattern $I_{1+2}(y)$ exhibits a striking interference pattern, whose periodicity matches the de Broglie wavelength

$$\lambda = \frac{2\pi\hbar}{p}. \tag{9}$$

This confirms that electrons behave as waves under suitable conditions.

The result is extraordinary. Each electron still behaves like a localized particle when detected—it hits a single detector, transferring all its charge, energy, and momentum there.



FIG. 4. Bottom: With both slits S_1 and S_2 open, the histogram $I_{1+2}(y)$ shows interference fringes. Points like M correspond to constructive interference (enhanced counts), while points like X correspond to destructive interference (no counts).

Yet, when the experiment is repeated many times, the ensemble of detections builds up an interference pattern, indicating wave-like behavior.

At some points, such as M, the detected count rate is four times what it would be with one slit open, not just double. At others, such as X, no electrons arrive at all, even though both slits are open. This pattern is impossible to explain if electrons were classical particles following definite trajectories.

In Newtonian mechanics, a particle travels through one slit or the other, completely unaffected by whether the second slit exists. Hence, with both slits open, one would expect

$$I_{1+2} = I_1 + I_2, \tag{10}$$

a simple sum of intensities. But experimentally, $I_{1+2} \neq I_1 + I_2$; in fact, at certain points, the total intensity vanishes. This cannot be accounted for by any classical picture of particle motion.

Some might argue that electrons from one slit could collide with those from the other, altering the observed distribution. However, this explanation fails for two reasons. First, random collisions could not consistently produce such a beautifully structured and repeatable interference pattern. Second, and more decisively, the interference persists even when electrons are fired one at a time. At any moment, there is only a single electron in the apparatus and therefore there is no chance of collisions and yet, over time, the same interference pattern emerges.

This means each electron somehow "knows" about both slits and their separation. The only way this can happen is if the electron is accompanied by a wave: an extended entity

that passes through both slits simultaneously and interferes with itself. The pattern on the detector reflects the intensity of this underlying wave.

Thus, de Broglie's idea marks the end of the purely Newtonian view of particles. Matter, like light, exhibits both wave and particle aspects. The electron, though localized when detected, must be described by a wave that governs the probabilities of where it may be found.

III. PHOTOELECTRIC EFFECT AND COMPTON EFFECT

A. The Photoelectric Effect

One of the earliest clues to the particle nature of light came from the photoelectric effect. When light of frequency ν is incident on a clean metal surface, electrons are ejected from it. Classical wave theory predicted that increasing the intensity of light should increase the energy of emitted electrons. However, experiments revealed that:

- No electrons are emitted below a certain cutoff frequency ν_0, regardless of intensity.
- The kinetic energy of emitted electrons depends only on the frequency, not on intensity.
- The emission is instantaneous once light of sufficient frequency is incident.

Einstein explained these results by proposing that light consists of discrete quanta, or photons, each carrying an energy

$$E = h\nu, \tag{11}$$

where h is Planck's constant. When a photon strikes the metal, it transfers its entire energy to a single electron. Part of this energy is used to overcome the work function W of the metal, and the rest appears as the kinetic energy of the ejected electron:

$$K_{\max} = h\nu - W. \tag{12}$$

At the threshold frequency ν_0, the electron just escapes with zero kinetic energy, so

$$h\nu_0 = W. \tag{13}$$

The linear dependence of $K_{\max}$ on ν provides direct experimental confirmation of the photon energy relation $E = h\nu$. This marked the first clear demonstration that light interacts with matter as if it were composed of particles.

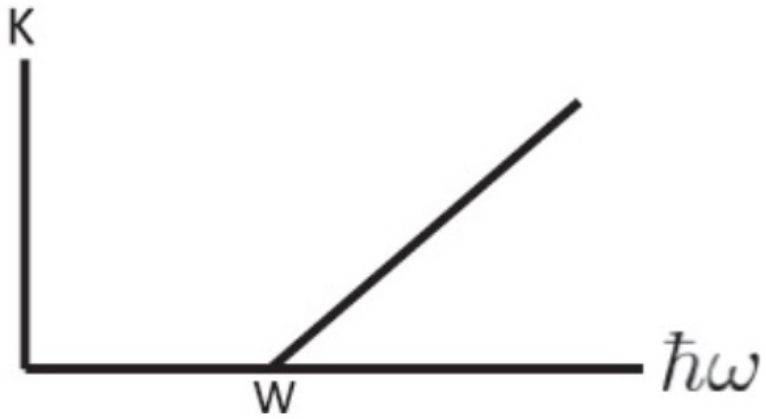


FIG. 5. Plot of Kinetic Energy versus Energy of incoming Photon in the Photoelectric effect.

B. The Compton Effect

One of the strongest confirmations of the photon picture of light came from the Compton effect, discovered by Arthur H. Compton in 1923. When X-rays are scattered by electrons in a material, the scattered radiation has a slightly longer wavelength than the incident one. The amount of wavelength shift depends on the scattering angle and cannot be explained by classical electromagnetic wave theory.

Compton explained this phenomenon by treating the X-ray photon as a particle with both energy and momentum. He imagined an elastic collision between a single photon and a nearly free electron at rest.

Let the incoming photon have wavelength λ, frequency $\nu = c/\lambda$, energy $E = h\nu = hc/\lambda$, and momentum $p = h/\lambda$. After the collision, the photon is scattered at an angle θ with a new wavelength λ', frequency $\nu' = c/\lambda'$, energy $E' = hc/\lambda'$, and momentum $p' = h/\lambda'$.

The initially stationary electron (mass m_e) recoils with velocity v and momentum $p_e = m_e v$ at an angle ϕ. We now apply conservation of energy and momentum to the collision.

1. Conservation of energy

Total energy before the collision:

$$E_{\text{initial}} = \frac{hc}{\lambda} + m_e c^2. \tag{14}$$

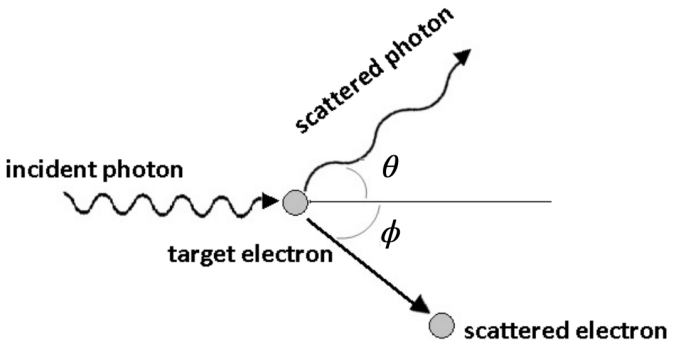


FIG. 6. Geometry of Compton scattering. An incident photon of wavelength λ collides with an electron at rest, scattering at an angle θ with a longer wavelength λ', while the electron recoils at angle ϕ.

Total energy after the collision:

$$E_{\text{final}} = \frac{hc}{\lambda'} + \sqrt{(p_e c)^2 + (m_e c^2)^2}. \tag{15}$$

Equating the two,

$$\frac{hc}{\lambda} + m_e c^2 = \frac{hc}{\lambda'} + \sqrt{(p_e c)^2 + (m_e c^2)^2}. \tag{16}$$

2. Conservation of momentum

In the x- and y-directions, taking the incident photon to move along the x-axis:

$$\frac{h}{\lambda} = \frac{h}{\lambda'}\cos\theta + p_e \cos\phi, \tag{17}$$

$$0 = \frac{h}{\lambda'}\sin\theta - p_e \sin\phi. \tag{18}$$

Squaring and adding Eqs. (17) and (18) to eliminate ϕ, we get

$$p_e^2 = \left(\frac{h}{\lambda}\right)^2 + \left(\frac{h}{\lambda'}\right)^2 - 2\frac{h^2}{\lambda\lambda'}\cos\theta. \tag{19}$$

Substitute Eq. (19) into the energy conservation equation (16). To simplify, isolate the kinetic energy of the electron:

$$\sqrt{(p_e c)^2 + (m_e c^2)^2} = m_e c^2 + \frac{hc}{\lambda} - \frac{hc}{\lambda'}. \tag{20}$$

Squaring both sides, we obtain

$$(p_e c)^2 + (m_e c^2)^2 = \left(m_e c^2 + \frac{hc}{\lambda} - \frac{hc}{\lambda'} \right)^2 . \tag{21}$$

Expanding the right-hand side,

$$\begin{aligned}(p_e c)^2 + (m_e c^2)^2 = m_e^2 c^4 + \frac{h^2 c^2}{\lambda^2} + \frac{h^2 c^2}{\lambda'^2} - \frac{2h^2 c^2}{\lambda\lambda'} \\ + \frac{2m_e hc^3}{\lambda} - \frac{2m_e hc^3}{\lambda'}.\end{aligned} \tag{22}$$

Cancelling $m_e^2 c^4$ from both sides gives

$$\begin{aligned}p_e^2 c^2 = h^2 c^2 \left(\frac{1}{\lambda^2} + \frac{1}{\lambda'^2} - \frac{2}{\lambda\lambda'} \right) \\ + 2m_e hc^3 \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right).\end{aligned} \tag{23}$$

From momentum conservation, Eq. (19), we have

$$p_e^2 = h^2 \left(\frac{1}{\lambda^2} + \frac{1}{\lambda'^2} - \frac{2\cos\theta}{\lambda\lambda'} \right). \tag{24}$$

Multiplying Eq. (24) by c^2 and substituting into Eq. (23), we get

$$\begin{aligned}h^2 c^2 \left(\frac{1}{\lambda^2} + \frac{1}{\lambda'^2} - \frac{2\cos\theta}{\lambda\lambda'} \right) \\ = h^2 c^2 \left(\frac{1}{\lambda^2} + \frac{1}{\lambda'^2} - \frac{2}{\lambda\lambda'} \right) \\ + 2m_e hc^3 \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right).\end{aligned} \tag{25}$$

Cancelling the common terms and rearranging, we obtain

$$\frac{h^2 c^2}{\lambda\lambda'}(1 - \cos\theta) = m_e hc^3 \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right). \tag{26}$$

Dividing both sides by $m_e hc^3$ and multiplying by $\lambda\lambda'$, we arrive at

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta). \tag{27}$$

Thus, the Compton wavelength shift is

$$\boxed{\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)}. \tag{28}$$

The quantity

$$\boxed{\lambda_C = \frac{h}{m_e c}} \tag{29}$$

is called the Compton wavelength of the electron, approximately $2.43 \times 10^{-12}\,\mathrm{m}$. ‘‘‘

The quantity

$$\lambda_C = \frac{h}{m_e c} \tag{30}$$

is called the *Compton wavelength* of the electron, with a value of approximately $2.43 \times 10^{-12}\,\mathrm{m}$. The shift $\lambda' - \lambda$ depends only on the scattering angle θ, not on the wavelength or intensity of the incoming X-rays.

At $\theta = 0°$, there is no shift; at $\theta = 180°$, the shift is maximum, equal to $2\lambda_C$. The remarkable agreement between Eq. (28) and experiment provided unambiguous evidence that photons carry both energy and momentum and behave like genuine particles during collisions with matter.

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[1] R. Shankar, *Fundamentals of Physics II: Electromagnetism, Optics, and Quantum Mechanics.*